

Notes on Bolton and Oehmke (2011): “Credit Default Swaps and the Empty Creditor Problem” *

Kostas Koufopoulos² and Giulio Trigilia³

²*University of York*

³*University of Rochester*

April 3, 2020

Abstract

This note shows that callable debt implements the efficient outcome in Bolton and Oehmke (2011), and it dominates any combination of debt and CDSs.

Key words: TBD

JEL classification: TBD

*All errors are our own.

Model. There are three dates $t = 0, 1, 2$, a firm and an outside financier. All agents are risk-neutral and consume at date two. The firm has no cash and needs to invest $F > 0$ at date zero, and generates two risky cash flows: $\tilde{C}_1$ at date one and $\tilde{C}_2$ at date two. Assume that $C_1 \in \{0, C_1^H\}$ and $\theta := \Pr.[C_1 = C_1^H] \in (0, 1)$. Also, assume that $C_2 \in \{C_2^L, C_2^H\}$ and $\phi := \Pr.[C_2 = C_2^H] \in (0, 1)$. Let $C_2^L > 0$. It is possible to liquidate the firm at date one, at a liquidation value normalized to zero. The following frictions are at play:

1. The date-one cash flow C_1 is not verifiable, and is privately observed by the firm;
2. The date-two cash flow is privately learned by the firm at the beginning of $t = 1$;
3. The date-two cash flow is verifiable at a cost $(1 - \lambda)C_2$, for $\lambda \in (0, 1)$, *only at $t = 1$* .

The authors assume that “at date 0 none of the date 2 cash flows can be contracted upon.” Which means, it is impossible to write a state-contingent division of cash flows at date two in the initial contract. However, they allow for credit default swaps to be issued, which are triggered by the verifiable event of renegotiation upon default at $t = 1$. We apply the same restriction here, which means that we allow for contracts to be contingent only on (i) the date 1 reported cash flows; and (ii) the event of renegotiation of the initial contract.

Consider the following callable debt contract. First, the financier writes a two-period simple debt contract with face value D . The contract gives the financier the right to collect $\min\{k, D\}$, where k denotes the amount of verifiable cash flows available at date 2, that is, $k \in \{0, C_2\}$. Specifically, $k = 0$ whenever there had been no verification at date one, and $k = C_2$ if verification did occur.¹ In addition, the debt contract specifies a call price P at which the debt can be bought back by the firm.

Suppose the firm reports the low cash flow at date one. There may be liquidation, but this is inefficient as it gives all parties a payoff of zero, as opposed to the date-two cash flow. If there is no verification, the financier knows that the entrepreneur will certainly default on the debt at date two, and $k = 0$ implies that the recovery rate from the debt would be zero. Therefore, the financier will certainly verify the date-two cash flow at time one, and at date two the financier will collect $\min\{D, C_2\}$.

Suppose the firm has a high cash flow at date one. One option is for the firm to call back the debt and obtain a payoff of either $C_1^H - P + C_2^H$ or $C_1^H - P + C_2^L$, depending on what the date-two cash flow will be (which the firm already knows at date one). Alternatively, the firm can restrain from calling the debt, in which case the financier will have to verify and, subsequently, collect again the $\min\{D, C_2\}$. Thus, the constraints required

¹Verification makes the payoff upon default a function of the date-two cash flow, but only indirectly.

for debt to be called are:

$$C_1^H - P + C_2^H \geq C_1^H + \lambda[C_2^H - D] \quad (IC_H)$$

$$C_1^H - P + C_2^L \geq C_1^H + \lambda[C_2^L - D]^+ \quad (IC_L)$$

The first IC constraint reflects the fact that the highest cash flow at date two is C_2^H , and so we must have $C_2^H \geq D$. IC_H can be written as $P \leq (1 - \lambda)C_2^H + \lambda D$, whereas IC_L can be written as $P \leq C_2^L - \lambda[C_2^L - D]^+$. Clearly, $C_2^L - \lambda[C_2^L - D]^+ < (1 - \lambda)C_2^H + \lambda D$, so IC_H is never binding.

Suppose that $C_2^L < D$. From the participation constraint of the financier we get:

$$\theta P + (1 - \theta)[\phi D + (1 - \phi)C_2^L] \geq F$$

Substituting $P = C_2^L$ and $C_2^H = D$ inside yields:

$$F \leq \theta C_2^L + (1 - \theta)\lambda[\phi C_2^H + (1 - \phi)C_2^L] \equiv \tilde{F}$$

Our $\tilde{F}$ is identical to the highest possible $\tilde{F}(q)$ in equation 11 of Bolton and Oehmke, which arises when $q = 1$.

It is still possible that the amount of financing needed exceeds F^* . In this case, there must be strategic default when $C_1 = C_1^H$ and $C_2 = C_2^L$. IC_L is violated and IC_H reads $P \leq C_2^H$. From the participation constraint for the financier we get:

$$F \leq \theta\phi C_2^H + \theta(1 - \phi)\lambda C_2^L + (1 - \theta)[\phi C_2^H + (1 - \phi)C_2^L] \equiv \tilde{F}'$$

Our $\tilde{F}'$ is equivalent to the highest possible $\tilde{F}'(q)$ defined in equation 12 of Bolton and Oehmke (which again arises when $q = 1$).

Thus, the callable debt contract we proposed is optimal, and it dominated any mixture of debt and CDSs, because of two reasons: (i) it eliminates the over-insurance problem; and (ii) it relaxes the financing constraints.